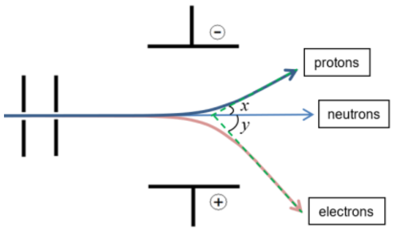
Deflection of particles



angle of deflection $\propto \left|\frac{q}{m}\right|$

In a particular experimental set-up, a beam of He^{2+} particles are deflected in an electric field by an angle of +10°. What angles will the beam of each of the following particles be deflected?

Li^{+}	
N^{3-}	

Proton no. & Nucleon no.

Subatomic particles

proton	relative mass: 1 relative charge: +1	${}^{1}_{1}p$
neutron	relative mass: 1 relative charge: 0	${}^{1}_{0}n$
electron	relative mass: $\frac{1}{1840}$ ≈ 0 relative charge: −1	${}^{0}_{-1}e$

Deduce the identity of X and Y

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{140}_{54}Xe + \mathbf{X} + 2\,{}^{1}_{0}n$$

$${}^{239}_{93}Np \rightarrow {}^{239}_{94}Pu + \mathbf{Y}$$

Orbital shapes

s - spherical - non-directional	1s	2s
p - dumb-bell - directional	$2p_x$	$2p_y$
	$2p_z$	$3p_z$
d	$3d_{xy}$	$3d_{yz}$
	$3d_{xz}$	$3d_{x^2-y^2}$
	$3d_{z^2}$	

Explaining Trends

Key points to explain atomic trends:

1. nuclear charge
2. shielding effect (no. of e⁻ shell & no. of e⁻)
3. electrostatic attraction between nucleus and outermost e⁻

atomic radii	• down the group: ____________ • across the period: ____________
ionic radii	• down the group: ____________ • across the period: - decreases across cations - sharp increase from cation to anion - decrease across anions
1st ionisation energy	• down the group: ____________ • across the period: ____________ **exception #1:** ns^2 vs $ns^2\,np^1$ s p s p p orbital is ________ in energy level compared to s orbital, hence less energy required to remove the electron in p orbital **exception #2:** $ns^2\,np^3$ vs $ns^2\,np^4$ s p s p ________ electron in p orbital experiences additional ________ repulsion, hence less energy is required to remove the electron
electro-negativity	• down the group: ____________ • across the period: ____________ *Note:* electrostatic attraction between nucleus and bonding e⁻
successive I.E.	• ____________

Electronic configuration, Electron-in-box & Energy level

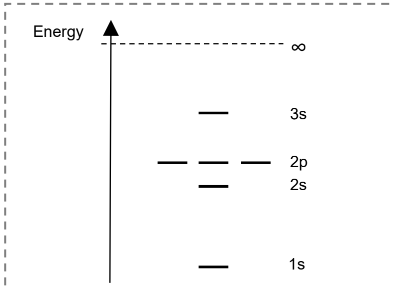
Method

- Order of filling up electrons:
  1s → 2s → 2p → 3s → 3p → __ → __ → 4p → 5s
- Order of writing electronic configuration:
  1s → 2s → 2p → 3s → 3p → __ → __ → 4p → 5s
- Orbitals can hold a maximum of two electrons and they must be of opposite spins.
- Orbitals must be occupied singly before pairing.
- Both Cr and Cu have anomalous electronic configuration.

Representations for O:

O	$1s^2$…

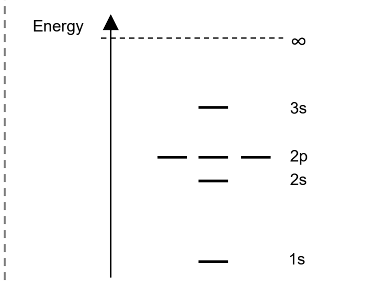
1s 2s 2p 3s



Representations for Na:

Na	

1s 2s 2p 3s



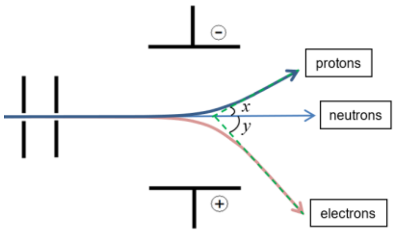
Extra Practice: Electronic Configuration

- State the electronic configuration of the following species

Fe	
Fe^{+}	
Fe^{3+}	

Note: For writing of electronic configuration of ions, <u>always start from the atom</u>

Deflection of particles



angle of deflection $\propto \left|\frac{q}{m}\right|$

In a particular experimental set-up, a beam of He^{2+} particles are deflected in an electric field by an angle of +10°. What angles will the beam of each of the following particles be deflected?

Li^+	For He^{2+} : +10° = k $(\frac{+2}{4})$ k = 20 For Li^+ : angle = 20 $(\frac{+1}{7})$ = +2.86°
N^{3-}	For N^{3-} : angle = 20 $(\frac{-3}{14})$ = − 4.29°

Proton no. & Nucleon no.

Subatomic particles

proton	relative mass: 1 relative charge: +1	${}^{1}_{1}p$
neutron	relative mass: 1 relative charge: 0	${}^{1}_{0}n$
electron	relative mass: $\frac{1}{1840}$ ≈ 0 relative charge: −1	${}^{0}_{-1}e$

Deduce the identity of X and Y

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{140}_{54}Xe + \mathbf{X} + 2\,{}^{1}_{0}n$$

${}^{a}_{b}\mathbf{X}$ 235 + 1 = 140 + a + 2(1) ⇒ a = 94
92 + 0 = 54 + b + 2(0) ⇒ b = 38

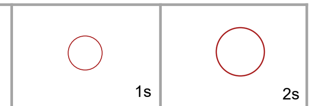
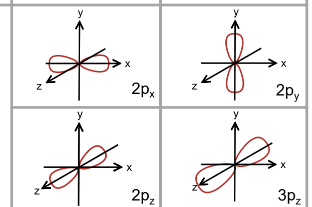
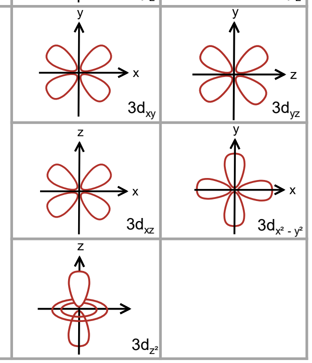
X is ${}^{94}_{38}Sr$

$${}^{239}_{93}Np \rightarrow {}^{239}_{94}Pu + \mathbf{Y}$$

${}^{a}_{b}\mathbf{Y}$ 239 = 239 + a ⇒ a = 0
93 = 94 + b ⇒ b = −1

Y is ${}^{0}_{-1}e^-$

Orbital shapes

s - spherical - non-directional	
p - dumb-bell - directional	
d	

Explaining Trends

Key points to explain atomic trends:

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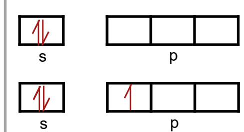
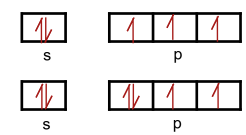
1st ionisation energy	• down the group: decreases • across the period: increases **exception #1:** ns^2 vs $ns^2\,np^1$ (see diagram) p orbital is higher in energy level compared to s orbital, hence less energy required to remove the electron in p orbital **exception #2:** $ns^2\,np^3$ vs $ns^2\,np^4$ (see diagram) paired electron in p orbital experiences additional electron-electron repulsion, hence less energy is required to remove the electron
electro-negativity	• down the group: decreases • across the period: increases *Note:* electrostatic attraction between nucleus and bonding e^-
successive I.E.	• increases





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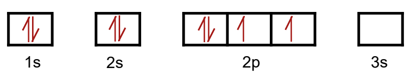
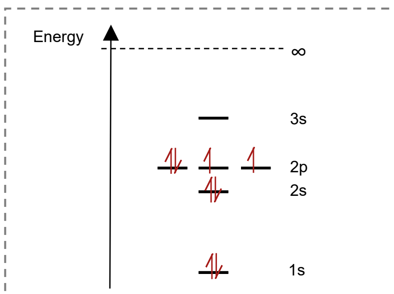
Representations for O:

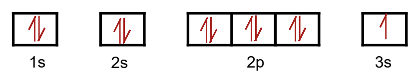
O	$1s^2\ 2s^2\ 2p^4$

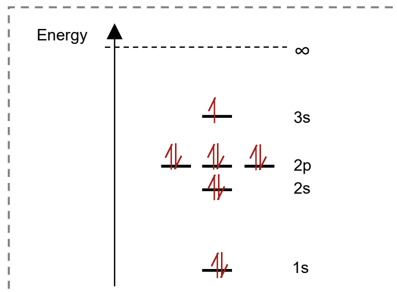




Representations for Na:

Na	$1s^2\ 2s^2\ 2p^6\ 3s^1$





Extra Practice: Electronic Configuration

- State the electronic configuration of the following species

Fe	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$
Fe^+	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^1$
Fe^{3+}	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$

Note: For writing of electronic configuration of ions, always start from the atom